Assignment 3
Algebra I (Monsoon-26)

September 23, 2026

1. Let n be a positive integer. Define the relation $\sim$ on $\mathbb{Z}$ by $a \sim b$ if and only if n divides $(a-b)$. This is an equivalence relation. Let $\mathbb{Z}/n\mathbb{Z}$ be the set of equivalence classes. Show that
$$\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \ldots, [n-1]\}.$$
Show that there is a unique law of composition
$$+_n : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$$
such that $\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is a morphism of additive magmas. Show that $(\mathbb{Z}/n\mathbb{Z}, +_n)$ is a group.

2. For any real numbers a and b with $a \neq 0$, define
$$f_{a,b} : \mathbb{R} \to \mathbb{R} \text{ by } f_{a,b}(x) = ax + b \text{ for all } x \in \mathbb{R},$$
let $G = \{f_{a,b} : 0 \neq a \in \mathbb{R} \text{ and } b \in \mathbb{R}\}$. Then prove that $(G, \circ)$ is a group, where $\circ$ is the composition of mappings.

3. For any rational numbers r and s, define $r \sim s$ if and only if $r - s$ is an integer. Then $\sim$ is an equivalence relation on the set $\mathbb{Q}$ of rational numbers. Let $\mathbb{Q}/\mathbb{Z}$ denote the set of equivalence classes w.r.t. $\sim$ in $\mathbb{Q}$. Show that there is a unique law of composition $+ : \mathbb{Q}/\mathbb{Z} \times \mathbb{Q}/\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ such that $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ $(x \mapsto [x]_\sim)$ is a morphism of additive magmas. Conclude that $(\mathbb{Q}/\mathbb{Z}, +)$ is a group.

4. Determine whether the set given is a subgroup of the group in each of the following.
   (i) $\mathbb{R}^+$ in $(\mathbb{R}, +)$, where $\mathbb{R}^+$ denote the set of all positive real numbers
   (ii) $\mathbb{R}^+$ in $(\mathbb{R} \setminus \{0\}, \cdot)$
   (iii) $\mathbb{Q}^+$ in $(\mathbb{R}^+, \cdot)$, where $\mathbb{Q}^+$ denote the set of all positive rational numbers
   (iv) $\pi\mathbb{Q}$ in $(\mathbb{R}, +)$, where π is an irrational number and $\pi\mathbb{Q} = \{\pi q \mid q \in \mathbb{Q}\}$

5. List all the elements in each of the following subgroups of the groups mentioned.
   (i) $\langle i \rangle$ in $(\mathbb{C} \setminus \{0\}, \cdot)$
   (ii) $\langle \sqrt{2} \rangle$ in $(\mathbb{R}^+, \cdot)$
   (iii) $\langle \sqrt{2} \rangle$ in $(\mathbb{R}, +)$

(iv) $\langle\sqrt{-2}\rangle$ in $(\mathbb{C}\setminus\{0\},\cdot)$

(v) $\langle 12\rangle$ in $(3\mathbb{Z},+)$.

6. Let $(S(X),\circ)$ be the group of all bijections of a nonempty set X onto itself and $x\neq y\in X$. Let
$$A_x=\{f\in S(X): f(x)=x\}$$
and
$$A_{x,y}=\{f\in S(X): f(x)=y\}.$$
Prove that A_x is a subgroup of $S(X)$ and that $A_{x,y}$ is not a subgroup of $S(X)$.

7. Determine which of the following are morphisms of groups between the given groups:

(i) Consider the group $(\mathbb{Z},+)$ and define $f:\mathbb{Z}\to\mathbb{Z}$ by $f(a)=2a$ for all $a\in\mathbb{Z}$.

(ii) Consider the group $(\mathbb{R},+)$ and define $f:\mathbb{R}\to\mathbb{R}$ by $f(x)=\cos x$ for all $x\in\mathbb{R}$.

(iii) Let Y be a nonempty subset of a set X and consider the group $P(X)$ and $P(Y)$ under the symmetric difference of sets. Define $f: P(X)\to P(Y)$ by $f(A)=A\cap Y$ for any $A\in P(X)$.

(iv) Let X and Y be as in (iii) above. Define $f:P(X)\to P(Y)$ by $f(A)=\mathcal{C}_Y(A)$ for any $A\in P(X)$.

(v) Let X be any nonempty set and $\mathbb{R}^X$ be the set of all mappings of X into $\mathbb{R}$. Consider the group $(\mathbb{R}^X,+)$ where $+$ is the point-wise addition. For any $x_0\in X$, define $f:\mathbb{R}^X\to\mathbb{R}$ by $f(\alpha)=\alpha(x_0)$ for all $\alpha\in\mathbb{R}^X$.

8. Consider the groups $(\mathbb{Z},+)$ and $(\mathbb{R},+)$ and, for any real number a, define $f_a:\mathbb{Z}\to\mathbb{R}$ by $f_a(x)=ax$ for all $x\in\mathbb{Z}$. Prove that a mapping $f:\mathbb{Z}\to\mathbb{R}$ is a morphism of groups if and only if $f=f_a$ for some $a\in\mathbb{R}$.

9. Let G be a group and $f:G\to G$ be defined by $f(x)=x^3$ for all $x\in G$. If f is an injective morphism of groups, then prove that G is commutative.

10. Let A, resp. B, be a set, endowed with an equivalence relation $\sim_A$, resp. $\sim_B$. Define a relation $\sim$ on $A\times B$ by setting
$$(a_1,b_1)\sim(a_2,b_2)\iff a_1\sim_A a_2 \text{ and } b_1\sim_B b_2.$$
(This is immediately seen to be an equivalence relation.)

(a) Use the universal property for quotients to establish that there are unique mappings
$$(A\times B)/\sim\to A/\sim_A$$
and
$$(A\times B)/\sim\to B/\sim_B$$
making the squares
$$\begin{array}{ccc} A\times B & \xrightarrow{\pi_1} & A \\ \downarrow & & \downarrow \\ (A\times B)/\sim & \longrightarrow & A/\sim_A \end{array} \qquad \begin{array}{ccc} A\times B & \xrightarrow{\pi_2} & B \\ \downarrow & & \downarrow \\ (A\times B)/\sim & \longrightarrow & B/\sim_B \end{array}$$
commute.

(b) Prove that $(A \times B)/\sim$, with these two functions, satisfies the universal property for the product of $A/\sim_A$ and $B/\sim_B$.

(c) Conclude that the resulting mapping

$$(A \times B)/\sim \to (A/\sim_A) \times (B/\sim_B)$$

is a bijection.

11. Prove that the groups $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \cdot)$ are isomorphic. Are the groups $(\mathbb{Q}, +)$ and $(\mathbb{Q}^+, \cdot)$ isomorphic?

12. Let A, B be sets and $F(A), F(B)$ the corresponding free groups. Assume $F(A) \cong F(B)$. If A is finite, prove that B is also and $A \cong B$.

13. Let F be the free group on $\{x, y\}$. Prove that the three elements $u = x^2$, $v = y^2$, and $z = xy$ generate a subgroup isomorphic to the free group on u, v, and z.

14. Prove that the group generated by x, y, z with the single relation $yxyz^{-2} = e$ is actually a free group.